

experiment_dataset_2021-09-23.zip

Dataset Description

Contact Ethan Prihar (ebprihar@wpi.edu) with questions.

Within experiment_dataset_2021-09-23.zip, there is a .zip file for each experiment. Within each experiment .zip file, there are four .csv files. Each .csv is described in its own section below.

exp_slogs.csv

This file contains one row for every action a student took within the ASSISTments tutor during their participation in the experiment.

experiment_id

The ID of the experimental problem set, which can be looked up using the ASSISTments website for reference.

student_id

A unique ID for the student participating in the experiment.

problem_id

If the action a student takes is related to a problem, this is the problem ID, which can be looked up using the ASSISTments website for reference.

problem_part

Some problems have multiple parts, this feature is an indicator of which part of the problem the student's action is related to.

scaffold_id

Some problems have sub-problems built in which break the problem down into smaller steps; these are called scaffolding problems. If a student's action is related to a scaffolding problem, this is the ID of that scaffolding problem.

experiment_tag_path

Within each experiment there are nested problem sets that students are randomly divided between. The names of these problem sets and the path a student takes through them can make it easier to determine what the student is experiencing (although they are mostly

indecipherable labels meant to help the researcher remember what conditions they've put in their study).

action

The specific action taken by the student.

timestamp

The date and time that the action was taken by the student.

assistments_reference_action_log_id

This ID corresponds to the row in the ASSISTments database that this information came from. It is possibly helpful for sorting the actions, but other than that, it is used as a reference ID for reviewing the dataset.

exp_plogs.csv

This file contains one row for every problem a student started within the ASSISTments tutor during their participation in the experiment.

experiment_id

The ID of the experimental problem set, which can be looked up using the ASSISTments website for reference.

student_id

A unique ID for the student participating in the experiment.

problem_id

If the action a student takes is related to a problem, this is the problem ID, which can be looked up using the ASSISTments website for reference.

problem_part

Some problems have multiple parts, this feature is an indicator of which part of the problem the student's action is related to.

scaffold_id

Some problems have sub-problems built in which break the problem down into smaller steps; these are called scaffolding problems. If a student's action is related to a scaffolding problem, this is the ID of that scaffolding problem.

problem_condition

This indicates which condition the problem is related to, this can be a control or treatment condition, pretest or posttest, or an unknown or ignored problem, which are excluded from further analysis.

start_time

The timestamp of when the student started the problem.

end_time

The timestamp of when the student finished the problem.

session_count

How many sessions the student took to complete the problem. Each time the student leaves the ASSISTments tutor and returns, the session count is incremented by one.

time_on_task

The total seconds, not including time the student was not using the ASSISTments tutor, that it took the student to complete the problem.

first_response_or_request_time

The time in seconds it took the student to make their first attempt to solve the problem or request tutoring.

first_answer

The student's first response to the problem. If the problem was an open-ended problem, the response will be "REDACTED" because the student could have written anything in their response and we want to preserve their anonymity.

correct

A binary indicator of whether or not the student got the problem correct on their first try with no tutoring.

reported_score

The score the student received for the problem. The score is a function of the number of attempts and the amount of tutoring the student used before correctly answering the problem.

answer_before_tutoring

A binary indicator of whether or not the student attempted to answer the problem before requesting tutoring.

attempt_count

How many attempts it took the student to answer the problem correctly.

hints_available

The number of hints available for the problem. Hints are short messages that help the student by prompting them with advice. They do not give away the answer.

hints_given

The number of hints the student viewed.

scaffold_problems_available

The number of scaffold problems available for the problem. Scaffold problems break the main problem down into smaller steps to help the student solve the main problem.

scaffold_problems_given

The number of scaffold problems the student was given.

explanation_available

A binary indicator of whether or not an explanation was available for the problem. An explanation is a large message that explains how to solve the problem and provides the answer.

explanation_given

A binary indicator of whether or not the student viewed an explanation.

answer_given

A binary indicator of whether or not the student was shown the answer.

assistments_reference_problem_log_id

This ID corresponds to the row in the ASSISTments database that this information came from. It is possibly helpful for sorting the problems, but other than that, it is used as a reference ID for reviewing the dataset.

exp_alogs.csv

This file contains one row for every student that participated in the experiment and contains information on how the student performed during the experiment.

experiment_id

The ID of the experimental problem set, which can be looked up using the ASSISTments website for reference.

student_id

A unique ID for the student participating in the experiment.

release_date

The date and time when the experimental assignment was assigned to the student.

due_date

The date and time when the experimental assignment was due for the student. This is blank if there was no due date.

start_time

The date and time when the experimental assignment was started by the student.

end_time

The date and time when the experimental assignment was completed by the student. This is blank if the student did not complete the assignment.

assignment_session_count

How many sessions the student took to complete the assignment. Each time the student leaves the ASSISTments tutor and returns, the session count is incremented by one.

assigned_condition

The experimental condition the student was assigned to.

condition_time_on_task

The average time it took a student to complete a problem for all the problems completed in condition.

condition_average_first_response_or_request_time

The average time in seconds it took the student to submit an answer or request tutoring for all the problems completed in condition.

condition_problem_count

The number of problems the student attempted in condition.

condition_total_correct

The number of problems the student got correct on their first try without tutoring in condition.

condition_total_correct_after_wrong_response

The number of problems the student got correct only after submitting an incorrect response in condition.

condition_total_correct_after_tutoring

The number of problems the student got correct only after requesting tutoring in condition.

condition_total_answers_before_tutoring

The number of problems the student attempted to answer before requesting tutoring in condition.

condition_total_attempt_count

The total number of attempts on all problems the student attempted in condition.

condition_total_hints_available

The total number of hints available on all problems the student attempted in condition.

condition_total_hints_given

The total number of hints viewed on all problems the student attempted in condition.

condition_total_scaffold_problems_available

The total number of scaffolding problems available on all problems the student attempted in condition.

condition_total_scaffold_problems_given

The total number of scaffolding problems given on all problems the student attempted in condition.

condition_total_explanations_available

The total number of explanations available on all problems the student attempted in condition.

condition_total_explanations_given

The total number of explanations viewed on all problems the student attempted in condition.

condition_total_answers_given

The total number of times the student was given the answer on all problems the student attempted in condition.

condition_session_count

How many sessions the student took to complete the experimental condition section of the assignment. Each time the student leaves the ASSISTments tutor and returns, the session count is incremented by one. This is blank if the student was never assigned to a condition.

assistments_reference_assignment_log_id

This ID corresponds to the row in the ASSISTments database that this information came from. It is possibly helpful for sorting the assignments, but other than that, it is used as a reference ID for reviewing the dataset.

The following five columns exist for both pretest and posttest. These columns will only have data if the experiment had the corresponding section and the student attempted to complete problems in that section.

*_problem_count

The number of problems the student attempted in this section.

*_correct

The number of problems the student got correct on their first try without tutoring in this section.

*_time_on_task

The average time it took a student to complete a problem for all the problems completed in this section.

*_average_first_response_time

The average time in seconds it took the student to submit an answer or request tutoring for all the problems completed in this section.

*_session_count

How many sessions the student took to complete the section of the assignment. Each time the student leaves the ASSISTments tutor and returns, the session count is incremented by one.

priors.csv

This file contains one row for every student that participated in the experiment and contains statistical and demographic information on the student, their class, their teacher, and their school district.

experiment_id

The ID of the experimental problem set, which can be looked up using the ASSISTments website for reference.

student_id

A unique ID for the student participating in the experiment.

student_prior_started_skill_builder_count

The number of skill builder assignments started by the student prior to the experiment. A skill builder is an assignment where a student must complete similar problems on a particular skill until they get three correct in a row.

student_prior_completed_skill_builder_count

The number of skill builder assignments completed by the student prior to the experiment.

student_prior_started_problem_set_count

The number of problem set assignments started by the student prior to the experiment. A problem set is an assignment where a student must complete all of a small set of problems that are typically part of a curriculum and related to the same set of skills.

student_prior_completed_problem_set_count

The number of problem set assignments completed by the student prior to the experiment.

student_prior_completed_problem_count

The number of problems completed by the student prior to the experiment.

student_prior_median_first_response_time

The median time in seconds for the student to attempt to answer or request tutoring on all their previously completed problems.

student_prior_median_time_on_task

The median time in seconds between a student starting and finishing all their previously completed problems.

student_prior_average_correctness

The fraction of times the student got a problem correct on their first attempt with no tutoring across all their previously completed problems.

student_prior_average_attempt_count

The average number of attempts for the student to complete a problem across all their previously completed problems.

class_id

A unique ID for the class of the student participating in the experiment.

class_creation_date

The date and time that the class was created in ASSISTments.

class_student_count

The number of students in the class.

class_prior_skill_builder_count

The number of skill builder assignments assigned to the class prior to the start of the experiment.

class_prior_problem_set_count

The number of problem set assignments assigned to the class prior to the start of the experiment.

class_prior_skill_builder_percent_started

The percent of assigned skill builder assignments started by members of the class prior to the start of the experiment.

class_prior_skill_builder_percent_completed

The percent of started skill builder assignments completed by members of the class prior to the start of the experiment.

class_prior_problem_set_percent_started

The percent of assigned problem set assignments started by members of the class prior to the start of the experiment.

class_prior_problem_set_percent_completed

The percent of started problem set assignments completed by members of the class prior to the start of the experiment.

class_prior_completed_problem_count

The number of problems completed by students in the class prior to the experiment.

class_prior_median_time_on_task

The median time in seconds between a student in the class starting and finishing all their previously completed problems.

class_prior_median_first_response_time

The median time in seconds for students in the class to attempt to answer or request tutoring on all their previously completed problems.

class_prior_average_correctness

The fraction of times the students in the class got a problem correct on their first attempt with no tutoring across all their previously completed problems.

class_prior_average_attempt_count

The average number of attempts for the students in the class to complete a problem across all their previously completed problems.

teacher_id

A unique ID for the teacher of the student participating in the experiment.

teacher_account_creation_date

The date and time the teacher created their ASSISTments account.

district_id

A unique ID for the school district of the student participating in the experiment.

location

The location of the school district

opportunity_zone

Whether or not the school district is located in an Opportunity Zone. Information on Opportunity Zones can be found here: <https://www.irs.gov/credits-deductions/businesses/opportunity-zones>.

locale_description

A brief description of the school district location.